Deepthi Venmanasseril Jainlal

AI /ML Engineer. Data Scientist· Human-AI Interaction · LLM Systems · Cloud-Native ML

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PROFESSIONAL SUMMARY

AI engineer and Master's student with 4+ years building and evaluating production AI systems — from multi-agent LLM pipelines to physical AI simulation and medical imaging deep learning. Excited about the intersection of AI capability and real-world usability: how systems behave when users interact with them, where they fail, and how prototyping can uncover new interaction patterns. Track record of translating complex AI architectures into interfaces and workflows that non-technical stakeholders can understand and act on. Background spans enterprise AI (Ford Motor Company, ~$23M business impact), safety-critical healthcare AI (Philips), and competitive hackathon prototyping (NVIDIA Cosmos, Stanford × DeepMind). Eager to bring a hands-on research and build mentality to the AIUX team at Google DeepMind.

CORE COMPETENCIES

AI Research & Prototyping LLM evaluation, experimental pipelines, rapid prototyping, multi-agent systems, physical AI simulation

Human-AI Interaction Stakeholder communication, AI output visualization, UX-adjacent design thinking, Streamlit/React/Angular UI

Evaluation & Measurement Model evaluation frameworks, multi-metric rubric design, A/B testing, synthetic dataset generation, ground-truth comparison

LLM & Agentic Systems LangChain, LangGraph, Haystack, FAISS, vLLM, Gemini API, NVIDIA Cosmos, RAG architectures, tool/function invocation

ML & Deep Learning CNNs, Transformers, PyTorch, TensorFlow/XLA, scikit-learn, YOLOv3, nnU-Net, CUDA GPU optimization

Cloud & Infrastructure GCP (BigQuery, Dataform, Cloud Storage), Docker, GitHub Actions CI/CD, MLflow, Databricks, NVIDIA H100 NVLink

PROFESSIONAL EXPERIENCE

Data Scientist – Supply Chain Intelligence · Ford Motor Company *Aug 2023 – Jul 2025 | Remote*

- Designed and shipped a production agentic AI system for structured intelligence extraction from enterprise documents — built the full lifecycle including RAG retrieval, orchestration, CI/CD, real-time observability, and executive-facing reporting; drove ~$23M in identified business savings.
- Built and iterated on interactive AI output dashboards (Power BI, Streamlit) and KPI reporting integrations that translated complex model behavior into actionable views for Finance, Engineering, and Procurement stakeholders — bridging the gap between AI capability and user understanding.
- Developed evaluation and data quality frameworks for production AI pipelines: statistical validation, anomaly detection, and signal fidelity monitoring; designed feedback loops that surfaced model failure modes to cross-functional teams.
- Engineered multi-variable predictive models on GCP/BigQuery/Dataform with abstraction layers enabling downstream analytics consumers; ~$5M in logistics savings.

ML Engineer / Software Engineer · Philips Healthcare *Jul 2021 – Jul 2023 | Remote*

- Built and deployed AI classification and risk analysis systems on clinical datasets in a safety-critical environment; designed full monitoring stack for streaming data pipelines — reduced manual analysis time by 60%.
- Developed Angular-based frontend features integrating AI/ML outputs into clinical tools used by domain experts; iterated based on feedback from clinicians on how outputs were interpreted and acted upon.
- Created automated QA frameworks with statistical validation, synthetic dataset generation, and ground-truth benchmarking for model evaluation in a regulated environment.

Deep Learning Research Intern · Philips Healthcare *Sep 2020 – Jul 2021 | India*

- Trained and optimized CNN-based object detection models (YOLOv3, nnU-Net) on 900+ annotated medical imaging datasets, reaching 89% detection accuracy; applied CUDA GPU optimization for scalable training pipelines.
- Defined repeatable model evaluation strategies against annotated ground truth; established test frameworks for measuring accuracy and performance in a regulated research context.

KEY PROJECTS

Physical AI Surgical Robot Simulation *· NVIDIA Cosmos Hackathon · Multi-Agent · H100 NVLink · vLLM · Gemini 2.5 Pro*

- Built a multi-agent motion planning and simulation pipeline on NVIDIA H100 NVLink using NVIDIA Cosmos Reason2 (8B via vLLM) and Gemini 2.5 Pro; designed Streamlit + React Three Fiber 3D UI for real-time interaction with the simulation — demonstrating end-to-end experimental prototyping from physical AI model to user-facing interface.
- Explored how users and operators interact with AI-generated surgical motion plans; iterated on interface design to surface model confidence, failure modes, and scenario outputs in an interpretable way.

MedMat – AI Decision-Support Platform *· Stanford × DeepMind Hackathon · Top 6 Finalist · Agentic AI · Regulatory Compliance*

- AI-native platform auto-generating structured regulatory compliance documentation using policy-based routing and agentic output generation; built evaluation layers, QA pipelines, and stakeholder reporting — Top 6 finalist at Stanford × DeepMind Hackathon.
- Designed UX flows enabling non-technical compliance teams to interact with and verify AI-generated outputs; demonstrated the challenge of making high-stakes AI decisions legible to domain experts.

Agentic Used-Car Inspection Agent *· LangGraph · Gemini 2.5 Flash · RAG · Computer Vision · Gradio*

- Autonomous inspection agent using computer vision, LLM reasoning, and RAG to identify vehicle defects and generate structured inspection reports; orchestrated via LangGraph with a Gradio interface for end-user interaction — built as capstone for Google AI Agents Intensive.

Supplier Intelligence Platform *· Ford Production · RAG · Agentic Pipeline · ~$23M Savings*

- End-to-end enterprise agentic AI: RAG retrieval, orchestration, structured extraction from unstructured documents, CI/CD deployment, real-time observability, and executive reporting integration.

Kaggle Lens Distortion Correction *· Computer Vision · Multi-Metric Evaluation · GPU-Accelerated*

- Classical and TF/XLA GPU-accelerated distortion correction pipeline; custom multi-metric rubric (Edge Similarity 40%, Line Straightness 22%, SSIM 15%) — demonstrates systematic AI performance measurement and evaluation framework design.

EDUCATION

MS, Engineering Management · Westcliff University, California · GPA: 4.0 *Oct 2025 – Oct 2027*

Business Analytics, Financial Management, Strategic Decision-Making

MTech, Computer Science · VIT University, India *Aug 2019 – Jul 2021*

Algorithms, Data Structures, Big Data, AI, Machine Learning, Statistics, Bayesian Inference